

Data Handoff — Ralph → Francesco

Project 11_02 — Near Real-Time Financial Forecasting using Amazon SageMaker

Overview

Ralph's job (steps 5–6) is to run sentiment analysis and generate forecasts. Once done, he passes the results to Francesco (steps 7–8) who stores them in DynamoDB and builds the QuickSight dashboard.

Ralph gives Francesco **two things**:

1. **Prediction results in S3** → for the historical charts on the dashboard
 2. **Latest predictions + anomaly flags in DynamoDB** → for the real-time status widget and Arthur's alerts
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Also: What Dalibor gives Ralph (for context)

Before Ralph can do anything, Dalibor gives him a **clean Parquet file in S3** — one per ticker.

S3 path:

```
s3://proj1102-data/forecast-ready/AAPL.parquet
s3://proj1102-data/forecast-ready/TSLA.parquet
```

File format:

Column	Type	Example	Notes
item_id	string	AAPL	Ticker name
timestamp	string	2026-04-18T10:00:00Z	UTC, ISO-8601, no gaps
target_value	float	185.42	The price to forecast
sentiment_score	float	0.87	From Comprehend (-1 to +1)

These column names are fixed. Amazon SageMaker is strict — any typo or rename will cause the import to fail.

1. What Ralph gives Francesco — S3 (for charts)

Amazon SageMaker automatically exports predictions as CSV files. Ralph configures the export path; Francesco points QuickSight at it via Athena.

S3 path:

```
s3://proj1102-data/forecast-output/AAPL/predictions_2026-04-18.csv
s3://proj1102-data/forecast-output/TSLA/predictions_2026-04-19.csv
```

CSV format (one row = one predicted time step):

Column	Type	Example	Notes
item_id	string	AAPL	Ticker
timestamp	string	2026-04-18T11:00:00Z	Predicted time
p10	float	183.10	Lower bound of confidence band
p50	float	185.42	Most likely predicted price
p90	float	187.80	Upper bound of confidence band

What Francesco does with this: He uses it to build the main chart on the QuickSight dashboard:

- Blue line → actual prices (from S3 raw data)
- Orange line → p50 (predicted price)
- Shaded area → p10 to p90 (confidence band — 90% of the time the real price lands here)

2. What Ralph gives Francesco — DynamoDB (for real-time widget + alerts)

For anything needing the **latest value right now** (current prediction, anomaly status), Ralph's Lambda writes directly to DynamoDB after each forecast cycle.

Table name: proj1102-forecasts

Schema:

Attribute	Key	Type	Example	Notes
ticker	Partition key	string	AAPL	One entry per ticker
timestamp	Sort key	string	2026-04-18T10:00:00Z	Time of prediction
p10	—	number	183.10	Lower bound
p50	—	number	185.42	Predicted price
p90	—	number	187.80	Upper bound
actual	—	number	192.10	Real price at that time
is_anomaly	—	boolean	true	Set by Ralph's Lambda
delta	—	number	+6.68	actual minus p50

Anomaly logic (Ralph's Lambda computes this):

```
if actual < p10 OR actual > p90:
    is_anomaly = True
else:
    is_anomaly = False
```

Example — normal case:

ticker	timestamp	p10	p50	p90	actual	is_anomaly	delta
AAPL	2026-04-18T10:00:00Z	183.10	185.42	187.80	185.90	false	+0.48

Actual (185.90) is inside the band → no anomaly.

Example — anomaly case:

ticker	timestamp	p10	p50	p90	actual	is_anomaly	delta
AAPL	2026-04-18T14:00:00Z	183.10	185.42	187.80	192.10	true	+6.68

Actual (192.10) is above p90 (187.80) → anomaly flagged → Arthur's Lambda reads this and sends the SNS alert.

Full data flow (summary)

Dalibor

```
--> S3: s3://proj1102-data/forecast-ready/AAPL.parquet
    (item_id, timestamp, target_value, sentiment_score)
    |
    v
    Ralph (Comprehend + SageMaker)
    |
    |--> S3: s3://proj1102-data/forecast-output/AAPL/
    |      (item_id, timestamp, p10, p50, p90)
    |      -> Francesco reads for QuickSight charts
    |
    +--> DynamoDB: proj1102-forecasts
          (ticker, timestamp, p10, p50, p90,
           actual, is_anomaly, delta)
          -> Francesco reads for real-time widget
          -> Arthur reads is_anomaly for SNS alerts
```

What Francesco can start building now (with dummy data)

Francesco does not need to wait for SageMaker to be fully set up. He can create the DynamoDB table with the schema above and insert a few dummy rows to start building his dashboard and test his queries right away.

Dummy rows to insert manually:

```
{
  "ticker": "AAPL",
  "timestamp": "2026-04-18T10:00:00Z",
  "p10": 183.10,
  "p50": 185.42,
  "p90": 187.80,
  "actual": 185.90,
  "is_anomaly": false,
  "delta": 0.48
}
```

```
{
  "ticker": "AAPL",
  "timestamp": "2026-04-18T14:00:00Z",
  "p10": 183.10,
  "p50": 185.42,
  "p90": 187.80,
  "actual": 192.10,
  "is_anomaly": true,
  "delta": 6.68
}
```

Things to freeze between Ralph and Francesco (agree before coding)

Item	Agreed value
S3 forecast output path	s3://proj1102-data/forecast-output/{ticker}/
DynamoDB table name	proj1102-forecasts
Partition key	ticker
Sort key	timestamp
Timestamp format	ISO-8601 UTC
Anomaly condition	actual < p10 OR actual > p90

Any change to the above must be announced to the group immediately — a renamed column or path breaks both Ralph's writer and Francesco's reader.